

altairsim — CP/M and FDOS on an iCOM FD3712/FD3812 8" floppy

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altairsim cpm22.toml

48K CP/M 2.2 v1.0
for iCOM FD3712 and Altair

A>DIR
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These machines have the **iCOM FD3712/FD3812 8"** floppy controller. They run CP/M in single density and in double density. They also run both revisions of iCOM's own FDOS disk operating system. Each machine boots when you start it. The startup of each machine file is **RUN F000** for CP/M and **RUN C000** for FDOS. On a real machine, you did the same: you examined the PROM address and pressed RUN.

The iCOM controller is a **command and handshake** controller. The Tarbell and VersaFloppy controllers shift bits. The iCOM controller buffers a whole sector instead. The processor moves the bytes through two ports (**C0h** and **C1h**). The disk driver of the operating system runs from a **boot PROM** in high memory: **F000** for CP/M and **C000** for FDOS.

The single-density **FD3712** disk is $77 \times 26 \times 128 = 256,256$ bytes. The double-density **FD3812** disk has mixed density. Track 0 is single density. Tracks 1 to 76 are double density. The disk is 509,184 bytes.

Press **Ctrl-E** (STOP) to go back to the monitor at any time. Type **RUN** to continue. CP/M gets **Ctrl-C**, because **Ctrl-C** is a warm boot in CP/M. The FDOS prompt is **!**. The two FDOS revisions have different commands. **FDOS-III** takes word commands: **LIST** shows the directory. The original **FDOS-I** takes single-letter directives: **L** shows the directory, **A** assembles, and **P** prints.

The files

File	What it is
<code>cpm22.toml</code>	FD3712 single-density CP/M 2.2 . <code>base = "icom"</code> plus the disk in drive 0.
<code>cpm22-3812.toml</code>	FD3812 double-density CP/M 2.23 (Lifeboat). Uses the FD3812 boot PROM and the double-density disk.
<code>fdos-iii.toml</code>	iCOM's own FDOS-III disk operating system. Uses the FDOS boot PROM at <code>C000</code> and boots to the <code>!</code> prompt.
<code>fdos-i.toml</code>	iCOM FDOS-I , the original revision. Uses the same <code>C000</code> PROM as FDOS-III. Adds a RAM board (<code>C500</code> to <code>FFFF</code>), because FDOS-I loads its resident executive into high memory. Boots to <code>!</code> . Use single-letter directives (<code>L</code> for the directory).
<code>CPM22v1.0-3712-48K.DSK</code>	The single-density CP/M system disk.
<code>CPM22-3812-48K.dsk</code>	The double-density CP/M system disk.
<code>FDOS-III-2SI0.DSK</code>	The FDOS-III system disk.
<code>FDOS-I-2SI0.DSK</code>	The FDOS-I system disk.
<code>R.COM</code> , <code>W.COM</code> , <code>HDIR.COM</code>	The host bridge utilities. They copy files between the host folder and a CP/M disk. The single-density CP/M disk already has them, and <code>cpm22.toml</code> adds the host bridge board that they need.

The disks have no undo. Drive 0 is mounted read/write, as on a real machine. The guest writes to the disk for each file you create. In a clone, `git checkout` puts the disk image back. In a copy that you received, nothing does. To test writes, copy the folder first. Or add `readonly = true` to the drive.

The boot PROMs are Mike Douglas disassemblies. The disk images are from deramp.com (iCOM Floppy Systems). The machines read and write existing disks. They cannot make a **blank** format. This is the same as the other controllers in this repository. All the disks here are already formatted.